KARATINA UNIVERSITY

UNIVERSITY EXAMINATIONS

2020/2021 ACADEMIC YEAR

SECOND YEAR **SECOND** SEMESTER

SUPPLEMENTARY/SPECIAL EXAMINATION

FOR THE DEGREE OF

BACHELOR OF SCIENCE IN COMPUTER SCIENCE

COURSE CODE: COM 220

COURSE TITLE: SOFTWARE ENGINEERING

DATE: 2 PM **TIME: 15TH SEPTEMBER 2022**

INSTRUCTION TO CANDIDATES

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Instructions: Answer ALL Questions in SECTION A and ANY THREE in SECTION B

SECTION A is COMPULSORY (Carries 31 Marks)

QUESTION 1: (COMPULSORY) 16 MARKS

a) Define the following terms as used in software engineering **(4 marks)**
   i. Software Engineering
   ii. Object Oriented Design
   iii. Risk Management
   iv. Prototype

b) State **TWO** reasons why softwares fail **(2 marks)**

c) State the phase in software development under which Cost Benefit Analysis performed. **(2 marks)**

d) State **TWO** functional and **TWO** non-functional requirements in a Library system that allows users to search articles, e-books and also download using a single interface to a number of databases of articles in different libraries. **(4 marks)**

e) For a system software to be a good software, it must exhibit some characteristics. Outline **FOUR** such characteristics **(4 Marks)**

QUESTION 2: (COMPULSORY) 15 MARKS

a) As a software engineer in an IT firm, you are expected to portray software engineering ethics. List **FOUR** examples of these engineering ethics. **(4 marks)**

b) Explain **TWO** kinds of software products. **(4 marks)**

c) Highlight **THREE** importances of software Engineering **(3 marks)**

d) Explain **TWO** benefits of using the Waterfall model in software engineering. **(4 marks)**

SECTION B (Carries 39 Marks)

QUESTION 3: (13 MARKS)

e) You have been hired as a software developer in a water development project in Kilifi county. The project will install sensors to monitor the water pressure levels as well as detect leakages. The system will be an integrated system comprising water management aspect. You and your team are developing the requirement specification document for the system. Agile software development method was adopted.

Required:

i. Highlight **SIX** quality attributes necessary for the documents of software specification. **(6 marks)**

ii. Explain **TWO** benefits of using the above methodology in software development. **(4 marks)**

iii. State **THREE** reasons for validation in software development process . **(3 marks)**

QUESTION 4: (13 MARKS)

a) Swift Tech company is a software development company that develops softwares to aid business in automating their processes. They have adopted the Structured design approach in their projects.

Required

i. Explain the structured design approach. **(1 marks)**

ii. Highlight **FOUR** benefits of using the design mentioned above. **(4 marks)**

iii. Highlight **THREE** unified modeling language(UML) suitable for the above design. **(6 marks)**

QUESTION 5: (13 MARKS)

Safaricom PLC hired you as software engineer to be part of a project team that will develop a software for use by the blind to access online materials such as journals, e-books and other useful material for their study.

a) Given the above information, required:

i. Highlight **SIX** types of risks that may be encountered during the project **(6 marks)**

ii. Explain how hardware and software reliability are related. **(3 marks)**

iii. Explain **TWO** stakeholders involved in the project. **(4 marks)**

QUESTION 6: (13 MARKS)

a) Describe test cases in software engineering. **(1 mark)**

b) Explain **THREE** benefits of using software metrics in software engineering. **(6 marks)**

c) A company has managed to develop a system for its day today's process automation. Before the system is released for use, there are several types of testing to be carried. Explain **THREE** such strategies. **(6 marks)**

QUESTION 7: (13 MARKS)

a) Explain the purpose of case tools. **(1 marks)**

b) With the aid of a diagram discuss six CASE repository functions . **(12 marks)**